Final Project Report: Churn Prediction & Data Drift

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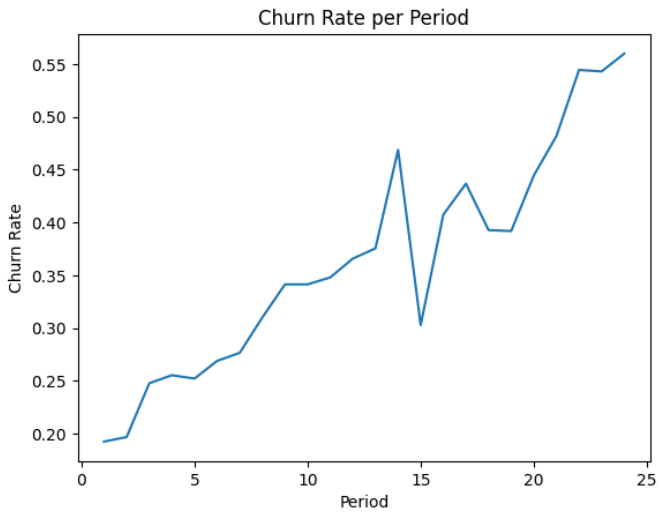
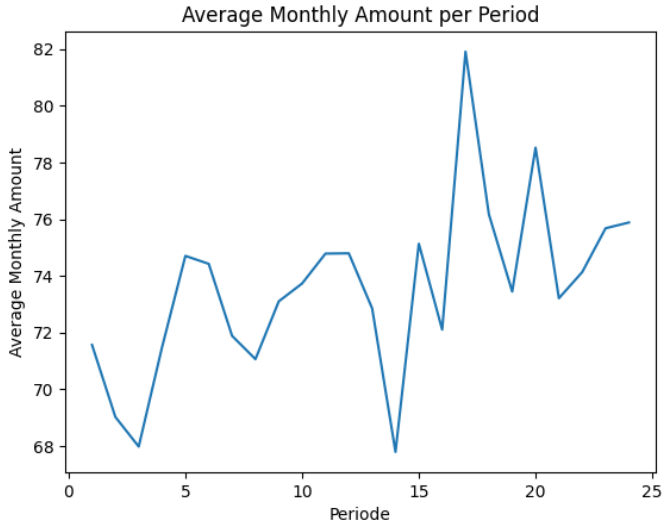
1. Drift-Oriented EDA

Our initial dataset analysis focused on detecting non-stationarity over time.Figure 1: Churn Rate per Period



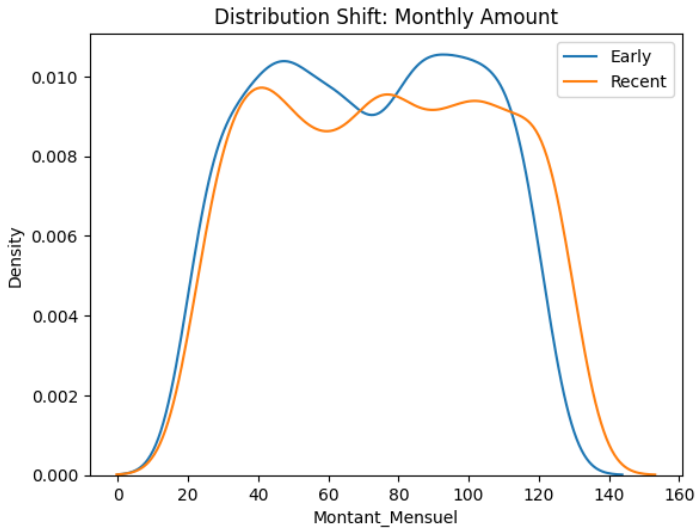
Analysis: We noticed that the number of people leaving isn't steady—it's actually increasing over time. This is called **Target Drift**. Because the "rules" of the business are shifting, a model trained on old data will eventually lose its touch. To stay accurate, our model needs to be updated regularly to keep up with these new trends.

Figure 2: Average Monthly Amount per Period



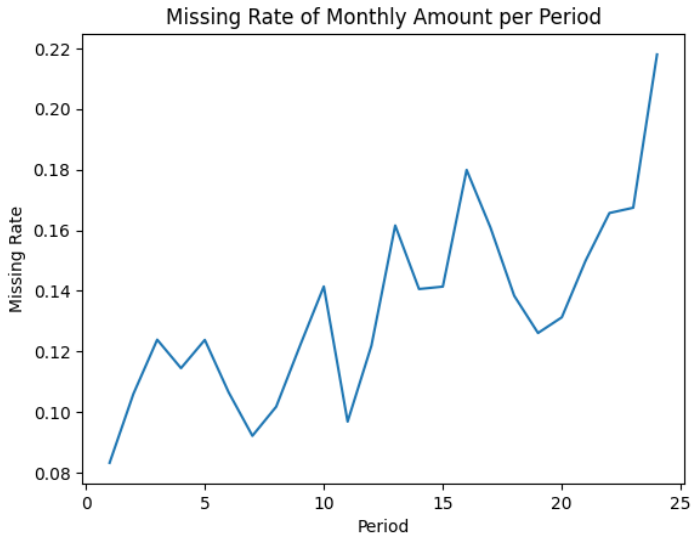
Analysis: We noticed that the average amount customers spend is changing over time. This is known as **Covariate Drift**. Because spending habits are evolving, we can't rely on a model that only understands how people spent money months ago. We need a system that can adapt to these new spending patterns as they happen.

Figure 3: Early vs Recent Distribution Comparison



Analysis: Comparing the density of *Montant_Mensuel* between the start and end of the study reveals a clear distribution shift. The "Recent" data shows a higher density at higher spend levels compared to the "Early" data, confirming that the model must handle evolving feature distributions to remain accurate.

Figure 4: Missing Values per Period



Analysis: We found that the amount of missing data isn't constant—it actually jumps from **8% to 20%** by the end of the study. This tells us the data is getting "messier" over time. Because these missing values are linked to customers leaving, we built the model to treat this "missingness" as a signal rather than just an error.

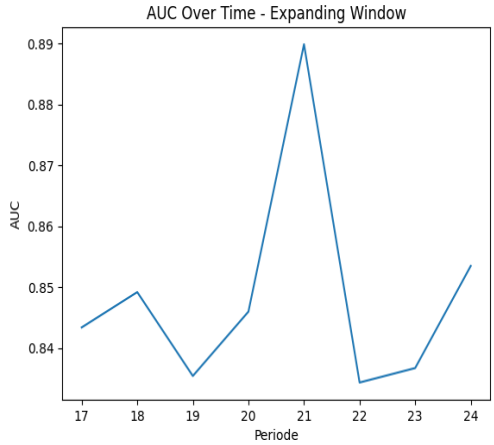
2. Pipeline Correctness

To ensure our results are honest, we built a secure data processing system that prevents any "cheating."

- The Process: We used a ColumnTransformer to prepare the data. It automatically fills in missing numbers using the middle value (Median) and converts categories into simple math (1s and 0s).
- The "No Cheating" Rule: Crucially, the pipeline only learns these rules from the past (Training Data). It never sees the "future" data during the setup phase.
- Why it matters: This stops Temporal Leakage, which happens when a model accidentally "peeks" at future trends to predict the past. By isolating the past from the future, we ensure our model's performance is realistic and reliable for real-world use.

3. Model Comparison

We evaluated Logistic Regression and Random Forest using an expanding window validation to simulate real-world time progression.



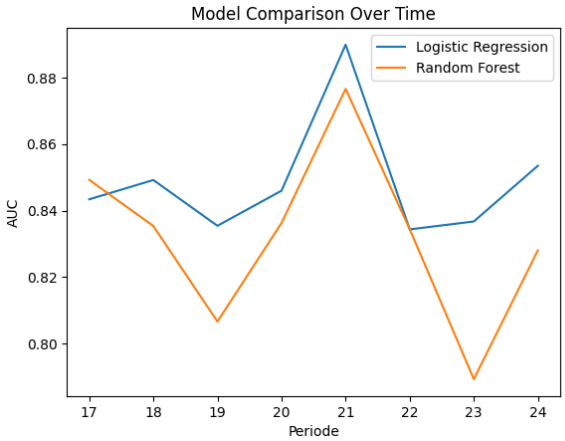


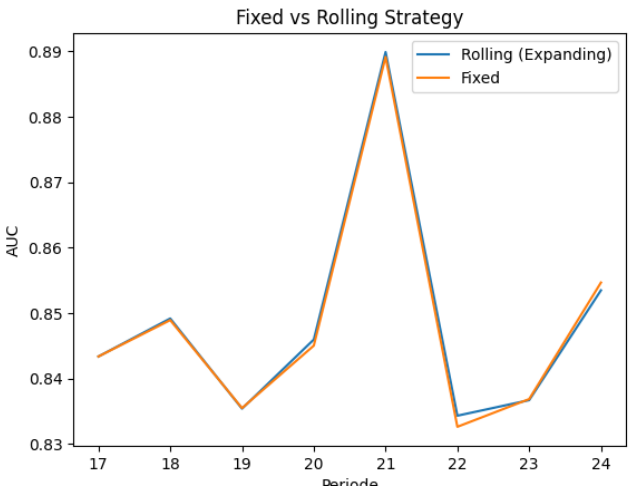
Figure Left: AUC Over Time – Logistic Regression
Figure Right: Model Comparison Over Time (Logistic vs RF)

Analysis: While the Random Forest model is theoretically better at capturing nonlinear patterns, it exhibited slightly higher variance across time periods. Conversely, Logistic Regression demonstrated much more stable performance over time, making it a more reliable baseline when dealing with shifting data

4. Retraining Strategy

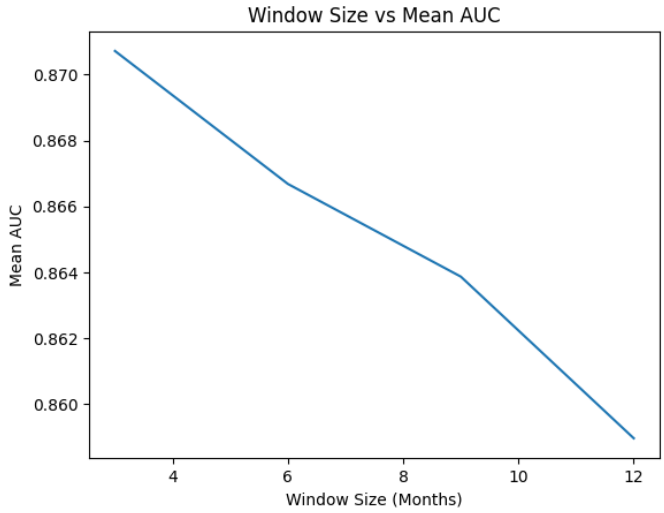
To combat the drift identified in our EDA, we tested two distinct retraining strategies: a Fixed strategy (trained once, never updated) and a Rolling strategy (continuously retrained on recent periods).

Figure: Fixed vs Rolling Strategy Comparison



Analysis: The empirical results clearly show that the rolling retraining strategy adapts better to data drift. The performance gap widens in the later periods, proving that continuously feeding the model recent data mitigates the degradation seen in the fixed model.

Figure: Window Size vs Mean AUC (Automatic Window Selection)



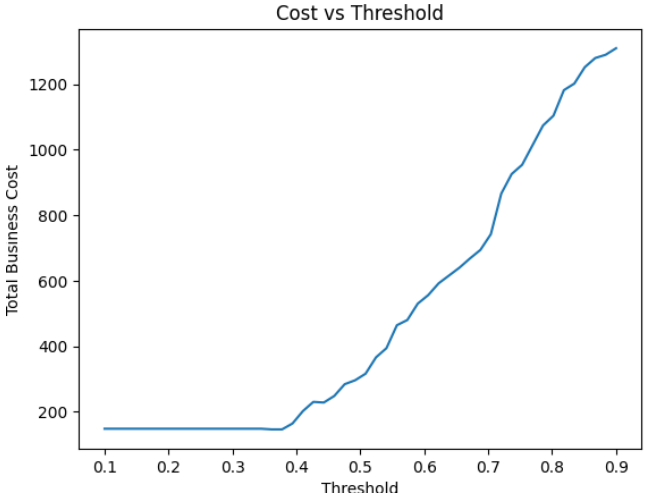
Analysis:
This graph answers one simple question: "**How much of the past should the model remember?"**

- The Test: We compared four "memory lengths" (3, 6, 9, and 12 months) to see which one predicted the future best.
- The Winner: The 3-month window was the clear winner with the highest score.
- The Reason: Since we already proved the data is changing (Drift), looking at data from a year ago is actually harmful. It’s like trying to predict what’s "trending" today by looking at what was popular last year.
- The Strategy: By using a short, 3-month "memory," the model stays focused on recent behavior, making it much faster at catching new patterns.

5. Business Optimization

Standard AI scores don't always tell you how much money you are making or losing. To fix this, we stopped focusing on "Accuracy" and started focusing on **Cost**.

Figure: Cost vs Threshold Curve



Analysis: By defining specific financial costs for false positives (unnecessary retention spend) and false negatives (lost customer revenue), we shifted our goal from maximizing raw accuracy to minimizing total business cost. The curve above identifies the exact threshold where the business saves the most money.

6. Conclusion

This project successfully demonstrates that in time-dependent churn prediction systems, static models are insufficient. Active drift monitoring, dynamic retraining strategies, and business-cost-driven decision-making are critical for maintaining a robust and profitable predictive pipeline.

*Ultimately, we selected the **Logistic Regression** model with a **3-month rolling retraining window** for production due to its consistent performance and ability to adapt to the identified data drift.*